



SILENTSPEAK: A HYBRID 3D-CNN AND BILSTM ARCHITECTURE FOR VISUAL SPEECH RECOGNITION

Dissertation

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INTRODUCTION

- Lip reading is a process of interpreting speech by visually analyzing the movements of the face detection, speaker's lips, tongue, and jaw.
- It is a crucial communication skill for people with hearing impairments .
- Automated lip reading system and assistive technologies are being developed to improve communication for people who have difficulty in speaking.
- The project aims to develop a system for lip movement reading to assist individuals with hearing impairments

PROBLEM DOMAIN

- The problem domain of this research lies at enabling hearing-impaired or non-verbal individuals to communicate through visual speech cues.
- Assisting stroke or neurological patients in speech recovery by analyzing and interpreting lip movements.
- Leveraging deep learning-based visual speech recognition enables accurate and real-time conversion of silent video into text, powering assistive technologies for those with speech or hearing loss.

CHALLENGES

- **Data Challenges:**

- Feature extraction often depends on pre-trained models that may not be tailored to the unique characteristics of lip reading data.

- **Model Complexity:**

- The use of dimensionality reduction techniques can add to the model's complexity, potentially complicating the training and optimization process.

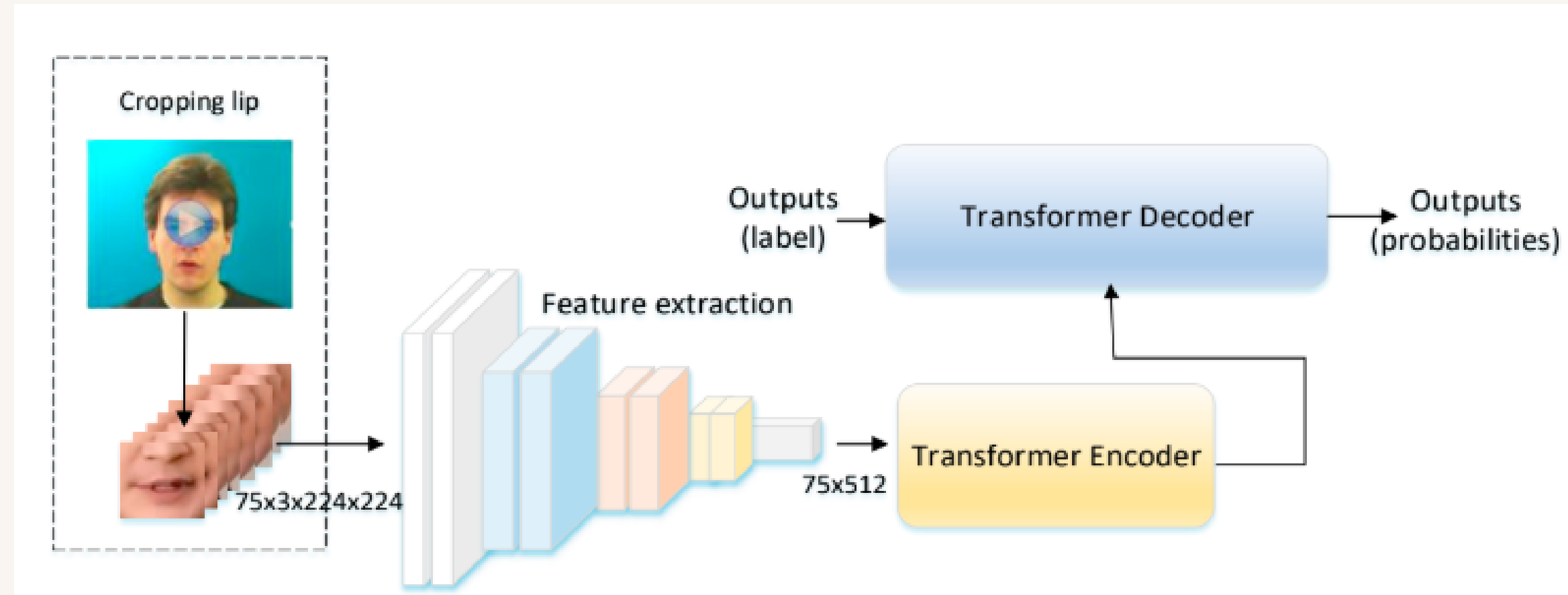
- **Evaluation Metrics:**

- Evaluation of lip reading systems should include not only word-level accuracy but also sentence-level and frame-level performance, along with robustness to noise and speaking style variations.

LITERATURE REVIEW

- **A Novel Machine Lip Reading Model**

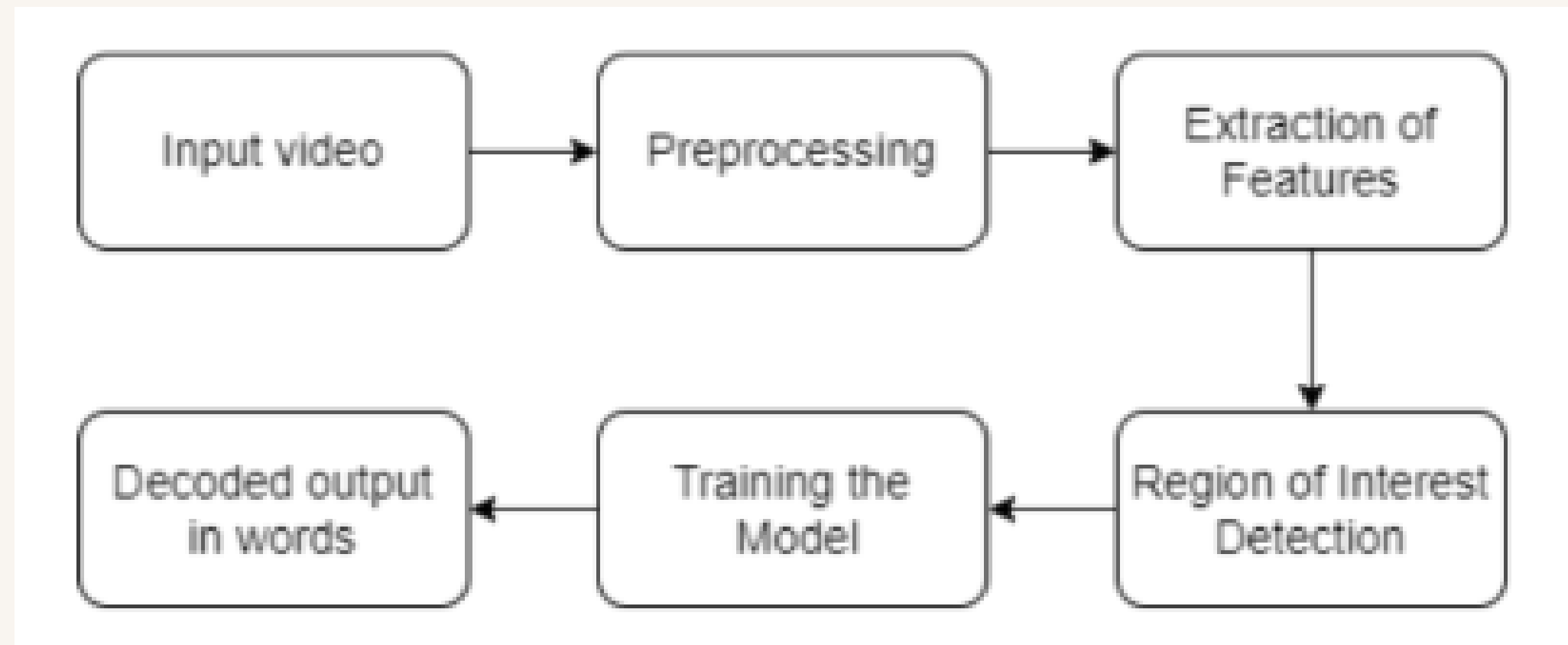
- Journal & Journal Name : The 8th International Conference on Information Technology and Quantitative Management (ITQM 2020 & 2021)
- Dataset Used : Grid Corpus
- Model :



- **Inference**

- The accuracy of our model reaches 45.81% in word-level lip reading, which forms a simple lip reading system.
- The recognition of the model still needs to be improved. I believe that after continuous debugging and
- Improvement of the model in the future, a lip reading model with high accuracy can finally be formed.

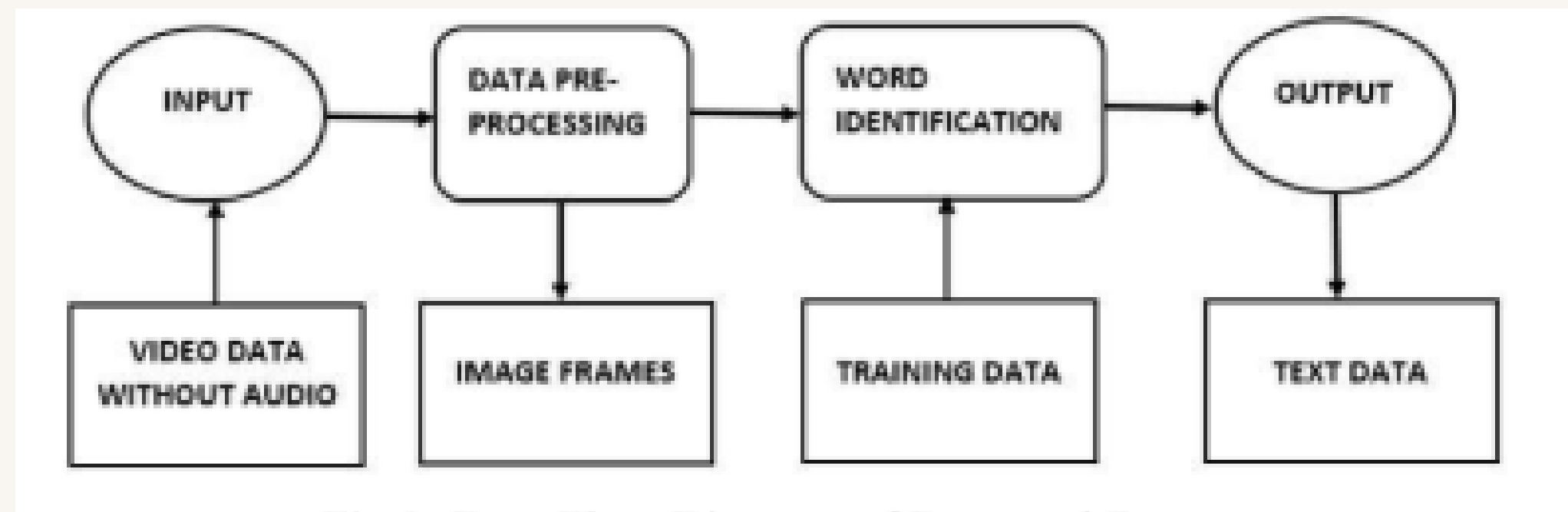
- **Text Extraction and Translation Through Lip Reading using Deep Learning (2024)**
 - Journal & Journal Name : (IJACSA) International Journal of Advanced Computer Science and Applications,
 - Dataset Used : LRW , LRW-1000.
 - Model :



- **Inference**

- The system leverages 2D CNNs, RNNs, and Transformer-based machine translation.
- Demonstrates exceptional accuracy in transcribing spoken words from lip movements.
- Improvement of the model in the future, a lip reading model with high accuracy can finally be formed.

- **Text Extraction through Video Lip Reading Using Deep Learning (2019 November)**
 - Journal & Journal Name : 8th International Conference on System Modeling & Advancement in Research Trends
 - Dataset Used :
 - Model :



- **Inference**

- Day by day the amount of video data are incrementing world wise. But the amount of systems that can analyze those data are not enough yet.
- They will understand each other in their own language and will be communicate in their native language.
- Using lip reading, voice data, and text analysis, it is possible to develop systems for seamless language translation.

MOTIVATION

- Accurate lip reading technology can greatly improve communication for people with hearing or speaking impairments, helping them participate more fully in daily life.
- Addressing the challenges of lip reading research will drive advancements in artificial intelligence, particularly in areas such as computer vision, natural language processing, and deep learning.
- Lip reading technology can be integrated into various human-computer interfaces, such as voice assistants and smart devices, enhancing their usability and accessibility for individuals with hearing impairments.

PROBLEM STATEMENT

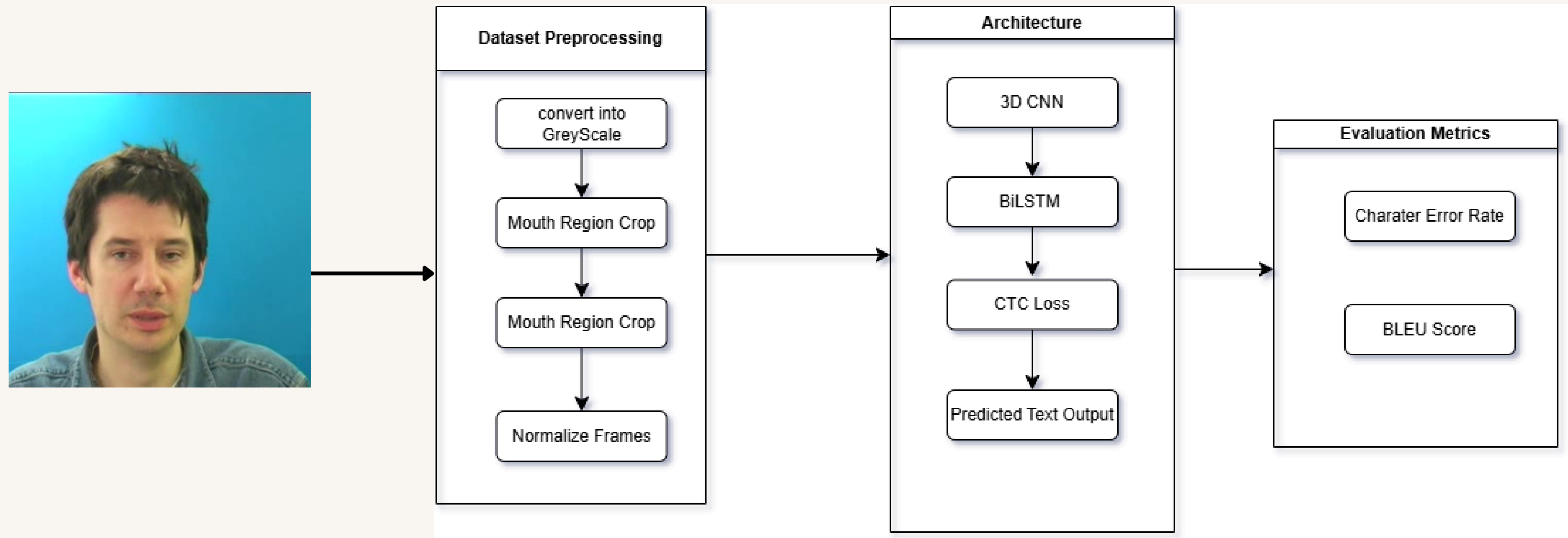
To design a machine learning architecture for lip reading that enhances the accuracy and reliability of visual speech recognition system by integrating robust feature extraction with temporal modeling techniques.

OBJECTIVES

- Identify the challenges and gaps in problem domain
- Understanding the problem domain by conducting literature survey
- Design an architecture for an effect machine learning model for solving the identified challenges in the problem domain
- Validate and fine tune the model
- Explore the possibilities to publish the new model.

METHODOLOGY

Architecture



DATASET



- Audio-Visual Dataset containing over 33,000 video clips of 34 speakers (18 male, 16 female).
- Each video features fixed-structure English sentences (e.g., "bin blue at J 2 now"), making it ideal for sequence learning.
- Provides high-quality, frontal face recordings, useful for lip movement analysis and visual speech recognition tasks.
- Includes both video and text alignment files to support supervised deep learning models like LipNet.

DATASET



Sample Image From Dataset

Phase	Speakers	Videos per Speaker	Total Samples
Phase 1	1	75	500
Phase 2	15	75	1125

Start (ms)	End (ms)	Word
0	23750	sil
23750	29500	bin
29500	34000	blue
34000	35500	at
35500	41000	f
41000	47250	two
47250	53000	now
53000	74500	sil

Sample of a align file

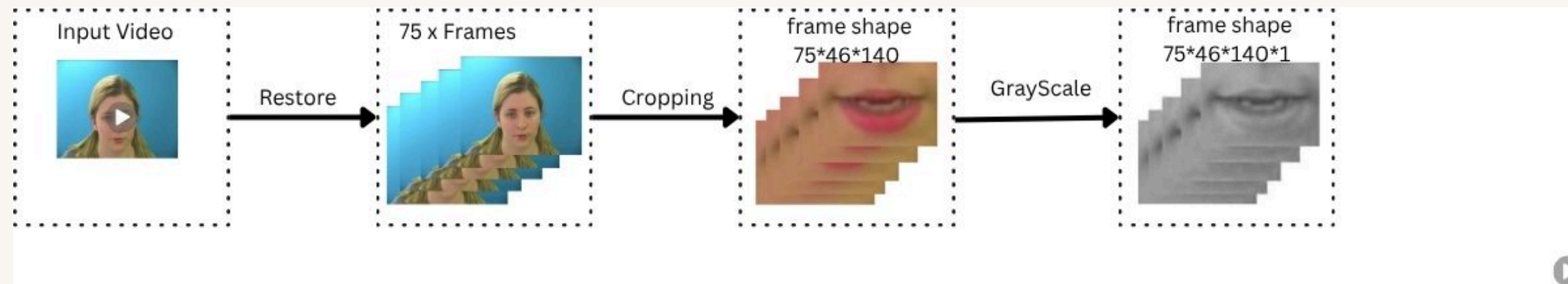
DATA PREPROCESSING

- Each video has 75 frames
- The longest sentence in the dataset contains 34 characters

```
100%|██████████| 1125/1125 [00:07<00:00, 150.53it/s]  
✓ MAX_FRAMES_PER_VIDEO: 75  
✓ MAX_LABEL_LENGTH: 34
```

DATA PREPROCESSING

- The input video is processed to extract 75 frames, capturing the temporal sequence of lip movements.
- Each frame is cropped to the mouth region, resulting in a shape of $75 \times 46 \times 140$.
- Cropped color frames are converted to grayscale, reducing dimensionality to $75 \times 46 \times 140 \times 1$ for efficient model input.

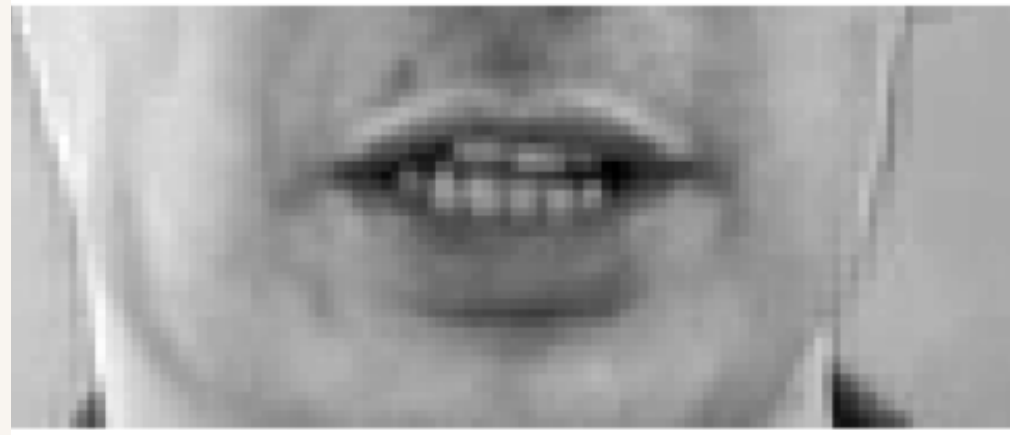


DATA PREPROCESSING

Frame 20



Frame 40



Middle Frame (37) — 'bin blue at I six again'

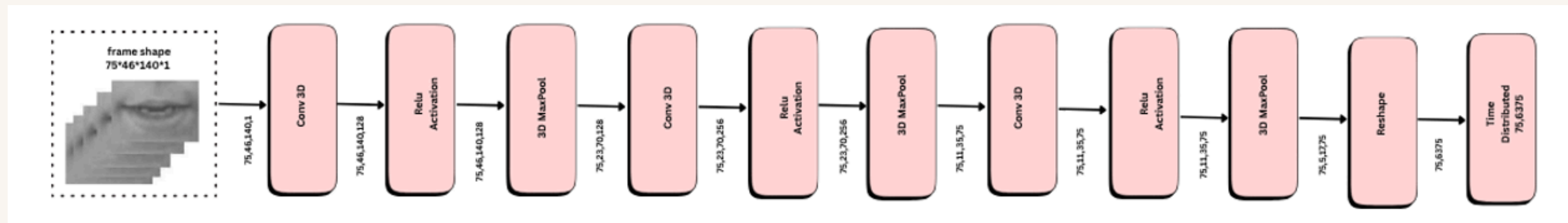


METHODOLOGY

- This section presents the deep learning architecture developed for visual speech recognition, specifically focusing on lip reading from silent video sequences.
- 3D-CNN extracts spatial and temporal features from consecutive mouth region frames.
- Flatten layer reshapes feature maps for sequential learning.
- BiLSTM captures context from both past and future frames for better prediction.
- CTC layer decodes unaligned frame-level features into text sequences.

3D CNN

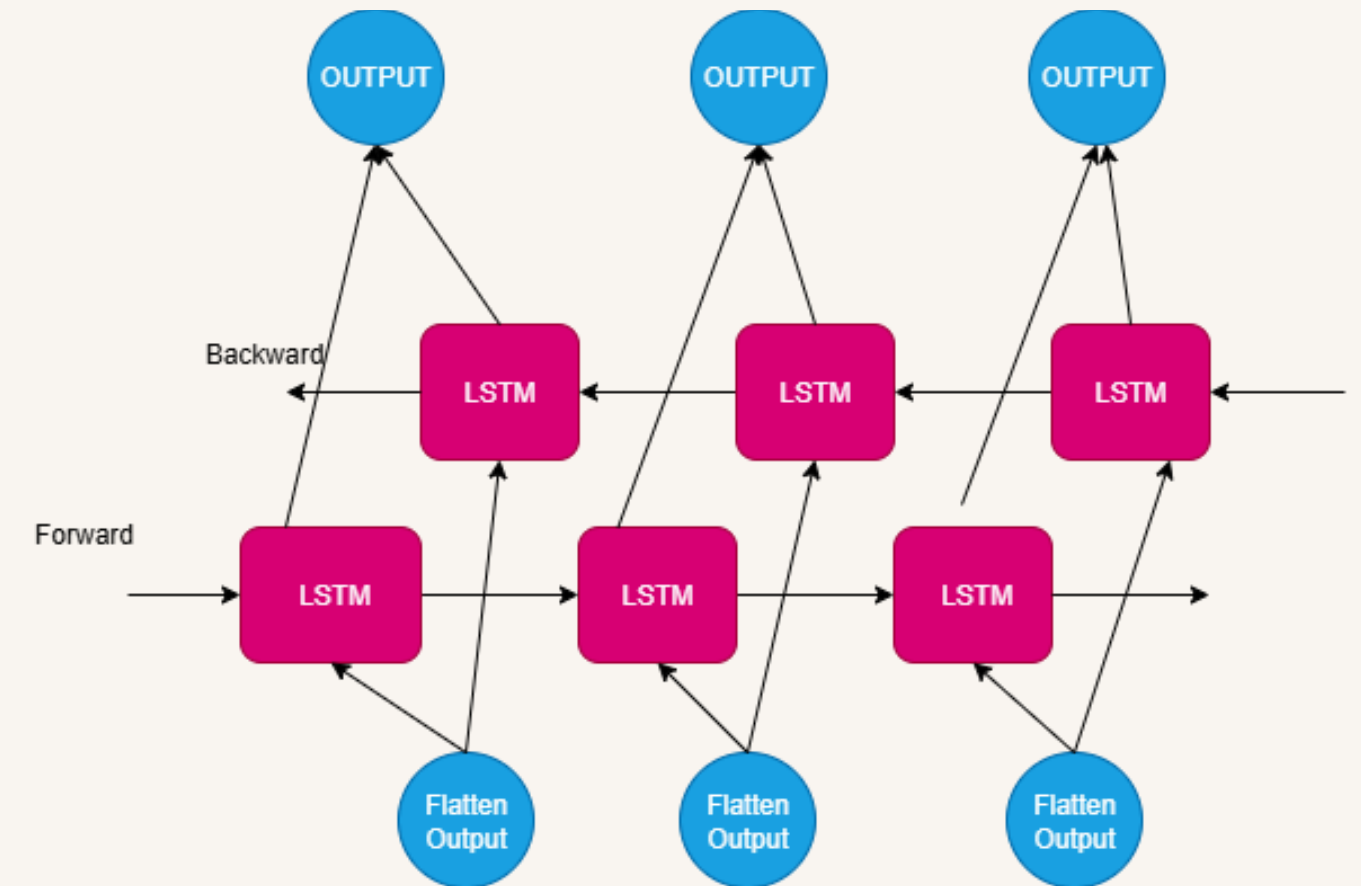
- The model takes input of shape $75 \times 46 \times 140 \times 1$, representing 75 grayscale video frames of size 46×140 pixels.
- The input passes through three Conv3D layers with 128, 256, and 75 filters, producing output shapes of $75 \times 46 \times 140 \times 128$, $75 \times 23 \times 70 \times 256$, and $75 \times 11 \times 35 \times 75$ respectively after pooling.
- After the final pooling, the output is reshaped to 75×6375 and passed through a TimeDistributed layer to retain temporal features.



3D CNN architecture

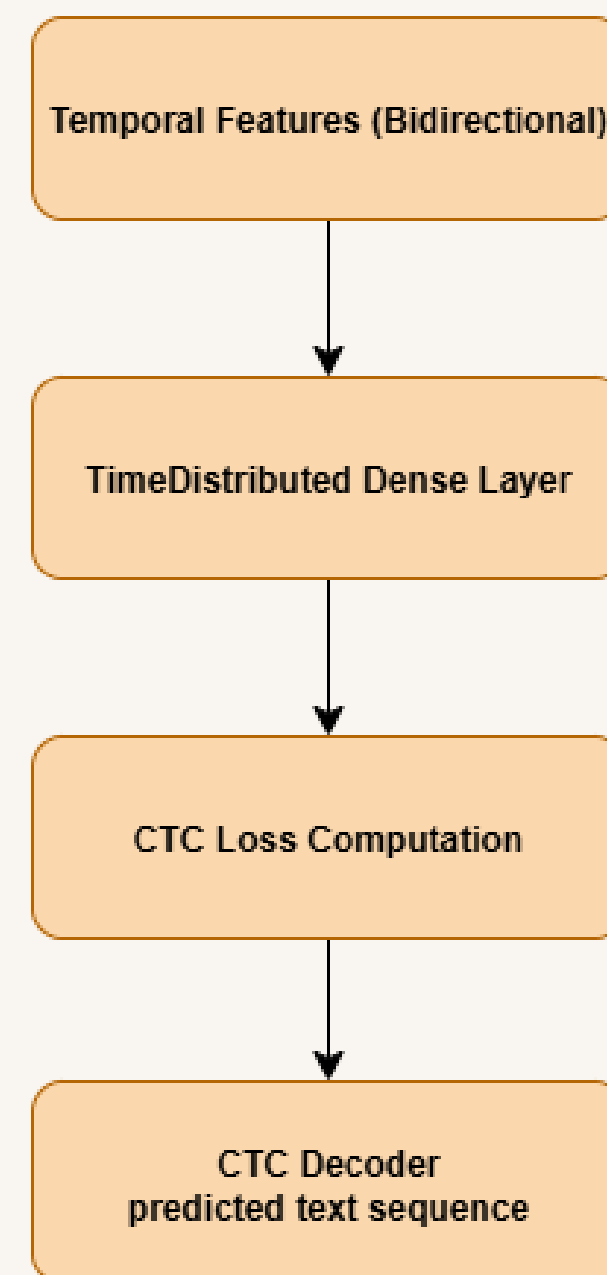
Bi-LSTM

- BiLSTM captures temporal dependencies by processing the input sequence forward and backward, improving context understanding from both past and future frames.
- It models how lips move over time, helping recognize word formations across the 75 time steps in the input.
- The outputs from both directions are concatenated to form a richer feature representation of shape per video sample.
- BiLSTM enhances lip-reading accuracy by aligning phonemes to characters using both past and future context.



Connectionist Temporal Classification Layer

- CTC handles unaligned input and output sequences, making it ideal for lip-reading tasks.
- It computes loss without requiring exact alignment between video frames and text labels.
- A TimeDistributed Dense layer generates character probabilities at each time step.
- The CTC decoder converts these probabilities into the final predicted text sequence.



RESULT

Epoch 1

```
... Epoch 1/50
1/1 ----- 1s 1s/step
Original: set red at b one soon
Prediction: le e e e e o
~~~~~
Original: place white with k seven again
Prediction: le e e e e o
~~~~~
450/450 ----- 924s 2s/step - loss: 99.8320 - val_loss: 70.0628 - learning_rate: 1.0000e-04
Epoch 2/50
1/1 ----- 0s 218ms/step
Original: set red with i four please
Prediction: la e e i o o
~~~~~
```

Epoch 100

```
450/450 ----- 434s 966ms/step - loss: 0.5857 - val_loss: 0.1269 - le
Epoch 100/100
450/450 ----- 0s 480ms/step - loss: 0.5030
Epoch 100: saving model to ./checkpoints/model_epoch_100_val_loss_0.3354.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras
1/1 ----- 0s 68ms/step
Original: set green at c three soon
Prediction: set gren at c thre son
~~~~~
Original: set blue at t six now
Prediction: set blue at t six now
~~~~~
450/450 ----- 434s 966ms/step - loss: 0.5031 - val_loss: 0.3354 - le
<keras.src.callbacks.history.History at 0x7b6f250223d0>
```

RESULT



Result Table - Phase 1 with 1 Speaker

Phase	Speakers	Average CER	Average BLEU
Phase 1	1	0.0021	0.403

Phase 2 Result

Epoch 9

```
Epoch 9/100
1000/1000 — 0s 233ms/step - loss: 39.2403
Epoch 9: saving model to /content/checkpoints/model_epoch_09_val_loss_37.0191.keras
1/1 — 1s 919ms/step
Original   : bin blue by f eight again
Prediction : bin blue a e

~~~~~

Original   : bin blue by c two now
Prediction : bin blue a e o

~~~~~

1000/1000 — 269s 263ms/step - loss: 39.2403 - val_loss: 37.0191 - learning_rate: 1.0000e-04
Epoch 10/100
1000/1000 — 0s 234ms/step - loss: 38.0379
Epoch 10: saving model to /content/checkpoints/model_epoch_10_val_loss_35.5910.keras
1/1 — 0s 60ms/step
Original   : bin green at n two please
Prediction : bin blue it e

~~~~~

Original   : bin blue in k two please
Prediction : bin blue it e

~~~~~

1000/1000 — 262s 262ms/step - loss: 38.0379 - val_loss: 35.5910 - learning_rate: 1.0000e-04
Epoch 11/100
1000/1000 — 0s 233ms/step - loss: 37.0113
Epoch 11: saving model to /content/checkpoints/model_epoch_11_val_loss_34.5088.keras
1/1 — 0s 61ms/step
Original   : bin blue with s four soon
Prediction : bin blue it e on

~~~~~

Original   : bin blue by r four please
Prediction : bin blue it e oon

~~~~~

1000/1000 — 261s 261ms/step - loss: 37.0113 - val_loss: 34.5088 - learning_rate: 1.0000e-04
Epoch 12/100
1000/1000 — 0s 234ms/step - loss: 36.0290
Epoch 12: saving model to /content/checkpoints/model_epoch_12_val_loss_33.8245.keras
1/1 — 0s 62ms/step
Original   : bin green at g zero again
Prediction : bin blue it e on

~~~~~

Original   : bin blue at l five soon
Prediction : bin blue it e o

~~~~~

1000/1000 — 262s 262ms/step - loss: 36.0291 - val_loss: 33.8245 - learning_rate: 1.0000e-04
Epoch 13/100
1000/1000 — 0s 233ms/step - loss: 35.2156
Epoch 13: saving model to /content/checkpoints/model_epoch_13_val_loss_32.6470.keras
1/1 — 0s 59ms/step
Original   : bin blue in y one soon
```

Epoch 130

```
Epoch 126/130
1/1 — 1s 1s/step
[Epoch 126] Original: bin green at n four please
[Epoch 126] Predicted: bin green at g four please

~~~~~

1000/1000 — 329s 326ms/step - loss: 5.7379 - val_loss: 8.2512 - learning_rate: 1.0000e-04
Epoch 127/130
1/1 — 0s 98ms/step
[Epoch 127] Original: bin blue in d seven soon
[Epoch 127] Predicted: bin blue in z teree soon

~~~~~

1000/1000 — 323s 321ms/step - loss: 6.1158 - val_loss: 13.5288 - learning_rate: 1.0000e-04
Epoch 128/130
1/1 — 0s 89ms/step
[Epoch 128] Original: bin blue by i six now
[Epoch 128] Predicted: bin blue in k six now

~~~~~

1000/1000 — 324s 322ms/step - loss: 8.8745 - val_loss: 10.8917 - learning_rate: 1.0000e-04
Epoch 129/130
1/1 — 0s 90ms/step
[Epoch 129] Original: bin blue by r nine soon
[Epoch 129] Predicted: bin blue by l nine soon

~~~~~

1000/1000 — 324s 322ms/step - loss: 9.2656 - val_loss: 8.6671 - learning_rate: 1.0000e-04
Epoch 130/130
1000/1000 — 0s 291ms/step - loss: 8.4521 📦 Downloading: /content/checkpoints/model_epoch_130_val_loss_9.5531.weights.h5
1/1 — 0s 89ms/step
[Epoch 130] Original: bin blue at i three soon
[Epoch 130] Predicted: bin blue at k thrhe soon

~~~~~

1000/1000 — 323s 321ms/step - loss: 8.4516 - val_loss: 9.5531 - learning_rate: 1.0000e-04
Restoring model weights from the end of the best epoch: 126.
<keras.src.callbacks.history.History at 0x7b7d30a4a890>
```

Epoch 100

```
1000/1000 — 527s 527ms/step - loss: 8.3863 - val_loss: 11.9354 - learning_rate: 1.0000e-04
Epoch 95/100
1000/1000 — 0s 488ms/step - loss: 6.5615 📦 Downloading: /content/checkpoints/model_epoch_95_val_loss_11.8257.weights.h5
1/1 — 0s 214ms/step
Original   : bin blue with e nine please
Prediction : bin blue at k five please

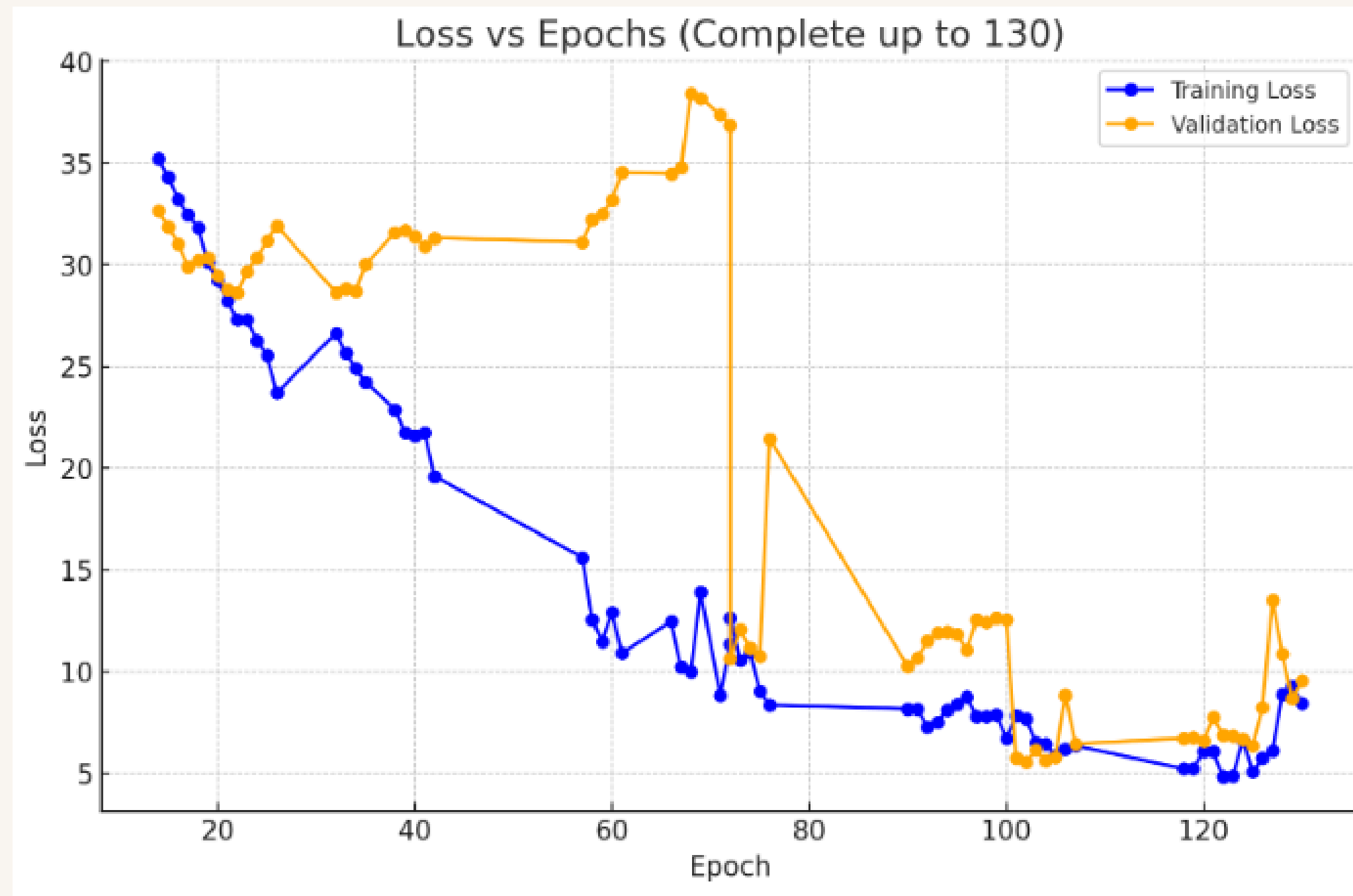
~~~~~

Original   : bin green at u one again
Prediction : bin green at g two nown

~~~~~

1000/1000 — 527s 527ms/step - loss: 6.5616 - val_loss: 11.8257 - learning_rate: 1.0000e-04
Epoch 95: early stopping
Restoring model weights from the end of the best epoch: 90.
<keras.src.callbacks.history.History at 0x7c7ad048ee90>
```

RESULT



Test Output

```
[Sample 1]
Ground Truth : bin blue at z six please
Prediction   : bin blue a e six please

[Sample 2]
Ground Truth : bin blue at e two sp soon
Prediction   : bin blue at e two p soon

[Sample 3]
Ground Truth : bin blue in e two please
Prediction   : bin blue in e two please

[Sample 4]
Ground Truth : bin blue by l four soon
Prediction   : bin blue by l four soon

[Sample 5]
Ground Truth : bin blue with s zero now
Prediction   : bin blue with z zoro now
```

```
[Sample 1]
Ground Truth : bin blue by e six please
Prediction   : bin blue by e six please

[Sample 2]
Ground Truth : bin blue by k nine please
Prediction   : bin blue by y nine please

[Sample 3]
Ground Truth : bin blue with l nine again
Prediction   : bin blue with l nine again

[Sample 4]
Ground Truth : bin blue at q nine now
Prediction   : bin blue at k nine now

[Sample 5]
Ground Truth : bin blue in u seven soon
Prediction   : bin blue in e seven soon
```


RESULT

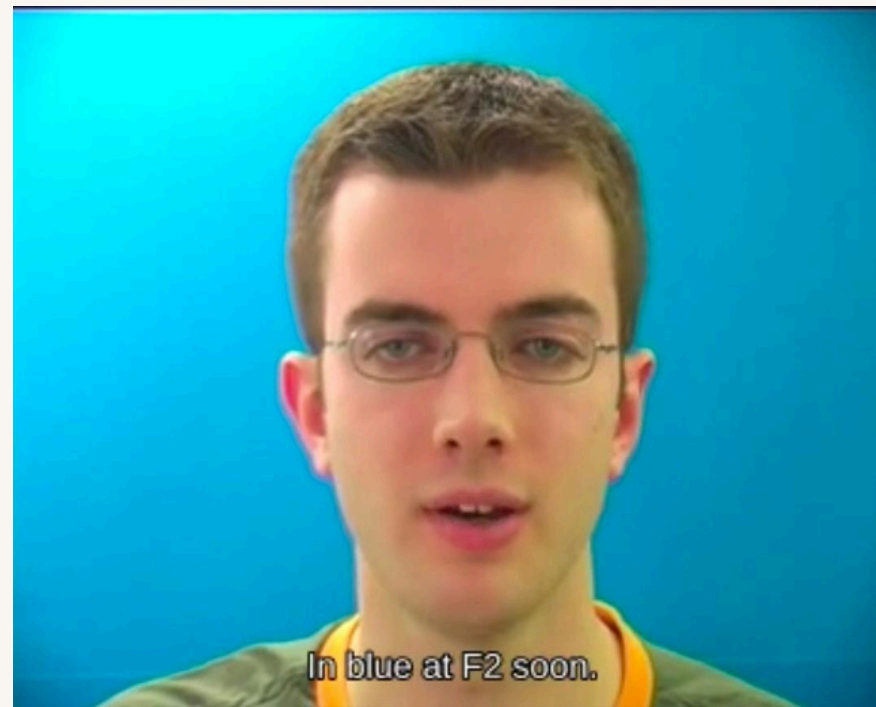
Samples 1-5

Sample	Ground Truth	Prediction	CER	WER	SER	BLEU SCORE
1	bin blue at z six please	bin blue a e six please	0.1053	0.3333	1	0.1374
2	bin blue at e two sp soon	bin blue at e two p soon	0.0526	0.1429	1	0.6435
3	bin blue in e two please	bin blue in e two please	0.0	0.0	0	1.0
4	bin blue by l four soon	bin blue by l four soon	0.0	0.0	0	1.0
5	bin blue with s zero now	bin blue with z zoro now	0.1053	0.3333	1	0.2512

Samples 6-10

Sample	Ground Truth	Prediction	CER	WER	SER	BLEU SCORE
1	bin blue by e six please	bin blue by e six please	0.0	0.0	0	1.0
2	bin blue by k nine please	bin blue by k nine please	0.0	0.0	0	1.0
3	bin blue with l nine again	bin blue with l nine again	0.0	0.0	0	1.0
4	bin blue at q nine now	bin blue at k nine now	0.0588	0.1667	1	0.2939
5	bin blue in u seven soon	bin blue in e seven soon	0.0526	0.1667	1	0.2939

Test Output



Performance Comparison phase 1 and Phase 2

Criteria	Single Speaker	15 Speakers
Dataset Size	450 training + 50 testing	1000+ training + 125 testing
Speaker Variability	Single person	Multiple (15 different speakers)
Average Character Error Rate (CER)	0.0021 (0.21%)	0.0295 (2.95%)
Word Error Rate (WER)	-	0.1011 (10.11%)
Sentence Error Rate (SER)	-	0.5040 (50.40%)
Max Frames per Video	75	75
Max Label Length	34	34

Evaluation Results on Full Test Set

CER: 0.0295

WER: 0.1011

SER: 0.5040

Lip Reading Models Comparisons

Model	Dataset	CER (%)	WER (%)	Reference
Proposed Model (15 Speakers)	GRID	2.95	10.11	—
Visual Speech Recognition with Cascaded Attention-CTC	GRID	3.2	6.2	Zhang et al. (2020) zhang2020spatio
LipNet: End-to-End Sentence-Level Lipreading	GRID	4.8	11.4	Assael et al. (2016) assael2016lipnet

CONCLUSION

- The hybrid 3D-CNN and BiLSTM model successfully captures spatiotemporal lip movement patterns for accurate visual speech recognition.
- The system achieves strong performance with a Character Error Rate of 2.95%, Word Error Rate of 10.11%, and Sentence Error Rate of 50.40%.
- The model generalizes well across different speakers and conditions, demonstrating its adaptability to unseen visual inputs.
- The results highlight the potential of deep learning in silent communication systems and open doors for further research and real-time applications.

FUTURE SCOPE

- Real-time webcam-based lip reading can be implemented using live video streaming for instant speech recognition.
- A hybrid system combining lip and audio input can improve accuracy for users with partial speech capability.
- The results highlight the potential of deep learning in silent communication systems and open doors for further research and real-time applications.
- Extending the model to support multiple languages like Malayalam, Hindi, and Tamil will increase its usability across diverse regions.

REFERENCES

- [1] H. Huang, C. Song, J. Ting, T. Tian, C. Hong, D. Zhang, and D. Gao, “A Novel Machine Lip Reading Model,” *Procedia Computer Science*, vol. 199, pp. 1432–1437, 2022.doi: 10.1016/j.procs.2022.01.181
- [2] S. T. K. Putcha, Y. S. V. Rajam, K. Sugamya, and S. Gopala, “Text extraction and translation through lip reading using deep learning,” *Int. J. Adv. Comput. Sci. Appl. (IJACSA)*, vol. 15, no. 6, 2024. DOI: 10./ijacsa.2024..
- [3] S. M. M. H. Chowdhury, M. Rahman, M. T. Oyshi, and M. A. Hasan, “Text Extraction through Video Lip Reading Using Deep Learning,” in *Proc. 8th Int. Conf. on System Modeling & Advancement in Research Trends (SMART)*, Moradabad, India, Nov. 2019, pp. 240–245. IEEE Conference ID: 46866. ISBN: 978-1-7281-3245-7.

THANK YOU